Luồng nghiệp vụ liên quan đến chương trình, giáo trình khóa học:

1. Subject Leader view list các subject đang có trong hệ thống:

```
SELECT s.id, s.code, s.name, s.description, s.status,
       ua.full_name AS created_by_name, s.created_at, s.updated_at
FROM subject s
LEFT JOIN user_account ua ON ua.id = s.created_by
ORDER BY s.created_at ASC
LIMIT 20 OFFSET 0;
```

Results Chart Export Source Primary Database Role postgres Run CTRL

id	code	name	description	status	created_by_name	created_at
2	JPN	Japanese	Japanese language training programs including JLPT preparation and c	active	Tanaka Yuki	2025-03-21 03:39:48.986989+00
1	ENG	English	English language training programs including IELTS, TOEIC, Business	active	Nguyen Thi English Leader	2025-02-19 03:39:48.986989+00

```
SELECT s.id, s.code, s.name, s.description, s.status,
       ua.full_name AS created_by_name, s.created_at, s.updated_at
FROM subject s
LEFT JOIN user_account ua ON ua.id = s.created_by
WHERE s.status = 'active'
  AND (s.code ILIKE '%eng%' OR s.name ILIKE '%eng%')
ORDER BY s.created_at ASC
LIMIT 20 OFFSET 0;
```

Results Chart Export Source Primary Database Role postgres Run selected CTRL

id	code	name	description	status	created_by_name	created_at
1	ENG	English	English language training programs including IELTS, TOEIC, Business	active	Nguyen Thi English Leader	2025-02-19 03:39:48.986989+00

Bước 1. Subject Leader tạo subject

- Validate quyền subject leader:

```
SELECT
  CASE WHEN EXISTS (
    SELECT 1
    FROM user_role ur
    JOIN role r ON r.id = ur.role_id
    WHERE ur.user_id = 6 AND r.code = 'SUBJECT_LEADER'
  )
  THEN 'User có quyền của Subject Leader'
  ELSE 'Lỗi: User không có quyền Subject Leader để tạo subject'
  END AS message;
```

Results Chart Export Source Primary Database Role postgres Run CTRL

message
User có quyền của Subject Leader

- Kiểm tra xem code subject mới có bị trùng không:

```
SELECT
  'CHI' AS code,
  CASE
    WHEN EXISTS (SELECT 1 FROM subject WHERE code = 'CHI')
      THEN 'Lỗi: subject code đã tồn tại'
    ELSE 'subject code chưa tồn tại'
  END AS message;
```

- Tạo Subject mới:

```
INSERT INTO subject (id, code, name, description, created_by, created_at)
VALUES (3,'CHI','Chinese','Chinese language programs and HSK exam preparation',6,
NOW())
RETURNING id, code, name, description, created_by,(SELECT ua.full_name FROM
user_account ua WHERE ua.id = subject.created_by) AS created_by_name, created_at;
```

Results Chart Export Source Primary Database Role postgres Run selected CTRL

id	code	name	description	created_by	created_by_name	created_at
3	CHI	Chinese	Chinese language programs and HSK exam preparation	6	Nguyen Thi English Leader	2025-10-27 17:06:06.936222+00

Bước 2. Subject Leader tạo Level cho subject

- Validate trùng code:

```
SELECT CASE
         WHEN EXISTS (
           SELECT 1
           FROM level l
           JOIN subject s ON s.id = l.subject_id
           WHERE s.code = 'CHI' AND l.code = 'HSK1'
         )
           THEN 'Lỗi: level code đã tồn tại trong subject này'
         ELSE 'OK'
       END AS message;
```

- Tạo level cho subject:

```
-- HSK1
INSERT INTO level (subject_id, code, name, standard_type, expected_duration_hours,
sort_order, description, created_at)
SELECT id, 'HSK1', 'HSK Level 1', 'HSK', 80, 1, 'Basic characters, simple
sentences', NOW()
FROM subject WHERE code = 'CHI'
ON CONFLICT (subject_id, code) DO NOTHING
RETURNING id, subject_id, code, name, standard_type, expected_duration_hours,
sort_order, description, created_at;

-- HSK2
```

```sql
INSERT INTO level (subject_id, code, name, standard_type, expected_duration_hours,
sort_order, description, created_at)
SELECT id, 'HSK2', 'HSK Level 2', 'HSK', 100, 2, 'Elementary grammar and
vocabulary', NOW()
FROM subject WHERE code = 'CHI'
ON CONFLICT (subject_id, code) DO NOTHING
RETURNING id, subject_id, code, name, standard_type, expected_duration_hours,
sort_order, description, created_at;

-- HSK3
INSERT INTO level (subject_id, code, name, standard_type, expected_duration_hours,
sort_order, description, created_at)
SELECT id, 'HSK3', 'HSK Level 3', 'HSK', 120, 3, 'Intermediate structures and
topics', NOW()
FROM subject WHERE code = 'CHI'
ON CONFLICT (subject_id, code) DO NOTHING
RETURNING id, subject_id, code, name, standard_type, expected_duration_hours,
sort_order, description, created_at

-- HSK4
INSERT INTO level (subject_id, code, name, standard_type, expected_duration_hours,
sort_order, description, created_at)
SELECT id, 'HSK4', 'HSK Level 4', 'HSK', 150, 4, 'Upper-intermediate grammar and
kanji/hanzi', NOW()
FROM subject WHERE code = 'CHI'
ON CONFLICT (subject_id, code) DO NOTHING
RETURNING id, subject_id, code, name, standard_type, expected_duration_hours,
sort_order, description, created_at;

-- HSK5
INSERT INTO level (subject_id, code, name, standard_type, expected_duration_hours,
sort_order, description, created_at)
SELECT id, 'HSK5', 'HSK Level 5', 'HSK', 180, 5, 'Advanced reading and writing',
NOW()
FROM subject WHERE code = 'CHI'
ON CONFLICT (subject_id, code) DO NOTHING
RETURNING id, subject_id, code, name, standard_type, expected_duration_hours,
sort_order, description, created_at;

-- HSK6
INSERT INTO level (subject_id, code, name, standard_type, expected_duration_hours,
sort_order, description, created_at)
SELECT id, 'HSK6', 'HSK Level 6', 'HSK', 200, 6, 'Proficiency for
academic/professional contexts', NOW()
```

```
FROM subject WHERE code = 'CHI'
ON CONFLICT (subject_id, code) DO NOTHING
RETURNING id, subject_id, code, name, standard_type, expected_duration_hours,
sort_order, description, created_at
```

Results Chart Export Source Primary Database Role postgres Run selected CTRL

id	subject_id	code	name	standard_type	expected_dura	sort_order	description	created_at
23	3	HSK6	HSK Level 6	HSK	200	6	Proficiency for academic/professional contexts	2025-10-27 17:30:10.305811+00

- Xem level trong subject:

```
SELECT
  l.id, l.subject_id, l.code, l.name,
  l.standard_type, l.expected_duration_hours, l.sort_order,
  l.description, l.created_at, l.updated_at
FROM level l
WHERE l.subject_id = 3
ORDER BY l.sort_order NULLS LAST, l.code;
```

Results Chart Export Source Primary Database Role postgres Run CTRL

id	subject_id	code	name	standard_type	expected_dura	sort_order	description	created_at	up
18	3	HSK1	HSK Level 1	HSK	80	1	Basic characters, simple sentences	2025-10-27 17:28:39.804519+00	20
19	3	HSK2	HSK Level 2	HSK	100	2	Elementary grammar and vocabulary	2025-10-27 17:29:39.174619+00	20
20	3	HSK3	HSK Level 3	HSK	120	3	Intermediate structures and topics	2025-10-27 17:29:43.638115+00	20
21	3	HSK4	HSK Level 4	HSK	150	4	Upper-intermediate grammar and kanji/hanzi	2025-10-27 17:29:47.877811+00	20
22	3	HSK5	HSK Level 5	HSK	180	5	Advanced reading and writing	2025-10-27 17:30:05.049657+00	20
23	3	HSK6	HSK Level 6	HSK	200	6	Proficiency for academic/professional contexts	2025-10-27 17:30:10.305811+00	20

Bước 3. Subject Leader define PLOs cho subject

- Validate:

```
SELECT CASE
        WHEN EXISTS (
          SELECT 1
          FROM plo p
          JOIN subject s ON s.id = p.subject_id
          WHERE s.code = 'CHI' AND p.code = 'PLO1'
        )
          THEN 'Lỗi: PLO code đã tồn tại trong subject này'
        ELSE 'OK'
      END AS message;
```

- Tạo PLOs:

```
INSERT INTO plo (subject_id, code, description, created_at)
SELECT s.id, v.code, v.description, NOW()
FROM subject s
CROSS JOIN (VALUES
  ('PLO1','Demonstrate comprehension of Chinese speech in common contexts'),
  ('PLO2','Understand Chinese written texts at intermediate level'),
  ('PLO3','Produce coherent written Chinese for daily and academic topics'),
  ('PLO4','Communicate orally in Chinese with acceptable fluency and accuracy')
```

```
) AS v(code, description)
WHERE s.code = 'CHI'
ON CONFLICT (subject_id, code) DO NOTHING
RETURNING id, subject_id, code, description
```

- Xem danh sách PLO trong subject:

```
SELECT
  p.id, p.code, p.description,
  p.created_at, p.updated_at
FROM plo p
WHERE p.subject_id = 3
ORDER BY p.code
```

Results Chart Export Source Primary Database Role postgres Run CTRL

id	code	description	created_at	updated_at
15	PLO1	Demonstrate comprehension of Chinese speech in common contexts	2025-10-27 19:19:25.897642+00	2025-10-27 19:19:25.897642+00
16	PLO2	Understand Chinese written texts at intermediate level	2025-10-27 19:19:25.897642+00	2025-10-27 19:19:25.897642+00
17	PLO3	Produce coherent written Chinese for daily and academic topics	2025-10-27 19:19:25.897642+00	2025-10-27 19:19:25.897642+00
18	PLO4	Communicate orally in Chinese with acceptable fluency and accuracy	2025-10-27 19:19:25.897642+00	2025-10-27 19:19:25.897642+00

- Tạo ra các course thuộc subject đó.

Bước 5. Tạo course

- Validate:

```
SELECT CASE
         WHEN EXISTS (
           SELECT 1
           FROM course c
           JOIN subject s ON s.id = c.subject_id
           WHERE s.code = 'CHI' AND c.logical_course_code = 'CHI-HSK3-PREP'
         )
           THEN 'Lỗi: Course code đã tồn tại trong subject này'
         ELSE 'OK'
       END AS message;
```

- Tạo course:

```
INSERT INTO course (subject_id, level_id, logical_course_code, version, code,
name, description, total_hours, duration_weeks, session_per_week,
hours_per_session, prerequisites, target_audience, teaching_methods,
effective_date, status, created_by, created_at)
SELECT
s.id, l.id, 'CHI-HSK3-PREP', 1, 'CHI-HSK3-PREP-V1', 'HSK3 Exam
Preparation','Comprehensive HSK3 prep: listening, reading, writing, speaking with
mock tests', 120, 12, 3, 2.5, 'HSK2 or equivalent','Learners targeting HSK3
certification', 'Task-based learning, drills, mock tests, feedback', CURRENT_DATE,
'draft', 6, NOW()
```

```
FROM subject s
JOIN level l ON l.subject_id = s.id AND l.code = 'HSK3'
WHERE s.code = 'CHI'
ON CONFLICT (code) DO NOTHING
RETURNING id, subject_id, level_id, logical_course_code, version, code, name,
description, total_hours, duration_weeks, session_per_week, hours_per_session,
prerequisites, target_audience, teaching_methods, effective_date, status,
created_by, (SELECT ua.full_name FROM user_account ua WHERE ua.id = created_by) AS
created_by_name, created_at;
```

Results Chart Export

id	subject_id	level_id	logical_course	version	code	name	description	tot
13	3	20	CHI-HSK3-PREP	1	CHI-HSK3-PREP-V1	HSK3 Exam Preparation	Comprehensive HSK3 prep: listening, reading, writing, speaking with	120

Results Chart Export

total_hours	duration_week	session_per_w	hours_per_ses	prerequisites	target_audience	teaching_methods	effective_d
120	12	3	2.50	HSK2 or equivalent	Learners targeting HSK3 certification	Task-based learning, drills, mock tests, feedback	2025-10-28

Results Chart Export

e	teaching_methods	effective_dat	status	created_by	created_by_name	created_at
ting HSK3 certification	Task-based learning, drills, mock tests, feedback	2025-10-28	draft	6	Nguyen Thi English Leader	2025-10-28 02:47:27.988938+00

Bước 6. Tạo CLOs

```
INSERT INTO clo (course_id, code, description, created_at)
SELECT id, 'CLO1', 'Apply listening strategies to understand HSK3 audio
materials', NOW()
FROM course WHERE code = 'CHI-HSK3-PREP-V1'
ON CONFLICT (course_id, code) DO NOTHING
RETURNING id, course_id, code, description;
```

Results Chart Export

id	course_id	code	description
17	13	CLO1	Apply listening strategies to understand HSK3 audio materials

- Xem danh sách CLO trong course

```
SELECT
  cl.id,
  cl.code,
  cl.description,
  cl.created_at,
  cl.updated_at
FROM course c
JOIN clo cl ON cl.course_id = c.id
WHERE c.code = 'CHI-HSK3-PREP-V1'
ORDER BY cl.code;
```

Results Chart Export Source Primary Database Role postgres Run CTRL

id	code	description	created_at	updated_at
17	CLO1	Apply listening strategies to understand HSK3 audio materials	2025-10-28 05:41:53.57697+00	2025-10-28 05:41:53.57697+00
18	CLO2	Interpret HSK3-level reading passages and answer comprehension quest	2025-10-28 05:42:20.456636+00	2025-10-28 05:42:20.456636+00
19	CLO3	Produce short written texts meeting HSK3 criteria	2025-10-28 05:42:24.115895+00	2025-10-28 05:42:24.115895+00
20	CLO4	Speak on everyday topics at HSK3 level with intelligibility	2025-10-28 05:42:39.743149+00	2025-10-28 05:42:39.743149+00

Bước 7. Mapping PLOs - CLOs

```
INSERT INTO plo_clo_mapping (plo_id, clo_id, status)
SELECT plo.id, clo.id, 'active'
FROM plo
JOIN subject ON subject.id = plo.subject_id AND subject.code = 'CHI'
JOIN course ON course.subject_id = subject.id AND course.code = 'CHI-HSK3-PREP-V1'
JOIN clo ON clo.course_id = course.id AND clo.code = 'CLO1'
WHERE plo.code = 'PLO1'
ON CONFLICT (plo_id, clo_id)
DO UPDATE SET status = 'active'
RETURNING
  (SELECT code FROM plo WHERE id = plo_clo_mapping.plo_id) plo_code,
  (SELECT code FROM clo WHERE id = plo_clo_mapping.clo_id) clo_code,
  status;
```

Results Chart Export Source Primary Database Role postgres Run CTRL

plo_code	clo_code	status
PLO1	CLO1	active

- Ma trận mapping của PLOs & CLOs

```
WITH data AS (
  SELECT
    clo.code AS clo_code,
    plo.code AS plo_code,
    CASE WHEN plo_clo_mapping.plo_id IS NOT NULL THEN '✓' ELSE '' END AS mark
  FROM subject
  JOIN course ON course.subject_id = subject.id
  JOIN clo ON clo.course_id = course.id
  JOIN plo ON plo.subject_id = subject.id
  LEFT JOIN plo_clo_mapping
    ON plo_clo_mapping.clo_id = clo.id
   AND plo_clo_mapping.plo_id = plo.id
   AND plo_clo_mapping.status = 'active'
  WHERE subject.code = 'CHI'
)
SELECT
  clo_code,
  MAX(CASE WHEN plo_code = 'PLO1' THEN mark END) "PLO1",
  MAX(CASE WHEN plo_code = 'PLO2' THEN mark END) "PLO2",
```

```
  MAX(CASE WHEN plo_code = 'PLO3' THEN mark END) "PLO3",
  MAX(CASE WHEN plo_code = 'PLO4' THEN mark END) "PLO4"
FROM data
GROUP BY clo_code
ORDER BY clo_code;
```

Results Chart Export Source Primary Database Role postgres Run CTRL

clo_code	PLO1	PLO2	PLO3	PLO4
CLO1	✓			
CLO2		✓		
CLO3			✓	
CLO4				✓

Bước 8. Check mapping CLOs & PLOs (xem còn CLO hay PLO nào đang chưa map hay không)

```
WITH clo_miss AS (
  SELECT clo.code
  FROM subject
  JOIN course ON course.subject_id = subject.id
  JOIN clo ON clo.course_id = course.id
  WHERE subject.code = 'CHI'
    AND NOT EXISTS (
      SELECT 1
      FROM plo_clo_mapping
      JOIN plo ON plo.id = plo_clo_mapping.plo_id
      WHERE plo_clo_mapping.clo_id = clo.id
        AND plo.subject_id = subject.id
        AND plo_clo_mapping.status = 'active'
    )
),
plo_miss AS (
  SELECT plo.code
  FROM subject
  JOIN plo ON plo.subject_id = subject.id
  WHERE subject.code = 'CHI'
    AND NOT EXISTS (
      SELECT 1
      FROM plo_clo_mapping
      JOIN clo ON clo.id = plo_clo_mapping.clo_id
      JOIN course ON course.id = clo.course_id
      WHERE plo_clo_mapping.plo_id = plo.id
        AND course.subject_id = subject.id
        AND plo_clo_mapping.status = 'active'
    )
```

```
)
SELECT
CASE
  WHEN (SELECT count(*) FROM clo_miss) = 0 AND (SELECT count(*) FROM plo_miss) = 0
    THEN 'Đã mapping đầy đủ (mọi CLO có ≥1 PLO và mọi PLO có ≥1 CLO)'
  WHEN (SELECT count(*) FROM clo_miss) > 0 AND (SELECT count(*) FROM plo_miss) > 0
    THEN 'Chưa mapping đủ: còn CLO và PLO chưa được map'
  WHEN (SELECT count(*) FROM clo_miss) > 0
    THEN 'Chưa mapping đủ: còn CLO chưa map PLO'
  ELSE
    'Chưa mapping đủ: còn PLO chưa map CLO'
END message;
```

Results Chart Export Source Primary Database Role postgres Run selected CTRL

message
Đã mapping đầy đủ (mọi CLO có ≥1 PLO và mọi PLO có ≥1 CLO)

Bước 9. Tạo Phase cho course

```
INSERT INTO course_phase (course_id, phase_number, name, duration_weeks,
learning_focus, sort_order, created_at)
SELECT id, 1, 'Skills Foundation', 4, 'Core grammar, vocabulary, listening/reading
basics', 1, NOW()
FROM course WHERE code = 'CHI-HSK3-PREP-V1'
ON CONFLICT (course_id, phase_number) DO NOTHING
RETURNING id, course_id, phase_number, name, duration_weeks, learning_focus,
sort_order, created_at;
INSERT INTO course_phase (course_id, phase_number, name, duration_weeks,
learning_focus, sort_order, created_at)
SELECT id, 2, 'Practice and Strategies', 4, 'Exam strategies, timed practice,
feedback', 2, NOW()
FROM course WHERE code = 'CHI-HSK3-PREP-V1'
ON CONFLICT (course_id, phase_number) DO NOTHING
RETURNING id, course_id, phase_number, name, duration_weeks, learning_focus,
sort_order, created_at;
INSERT INTO course_phase (course_id, phase_number, name, duration_weeks,
learning_focus, sort_order, created_at)
SELECT id, 3, 'Mock Tests and Review', 4, 'Full mock tests, targeted review,
consolidation', 3, NOW()
FROM course WHERE code = 'CHI-HSK3-PREP-V1'
ON CONFLICT (course_id, phase_number) DO NOTHING
RETURNING id, course_id, phase_number, name, duration_weeks, learning_focus,
sort_order, created_at;
```

Results Chart Export Source Primary Database Role postgres Run selected CTRL

id	course_id	phase_number	name	duration_week	learning_focus	sort_order	created_at
10	13	1	Skills Foundation	4	Core grammar, vocabulary, listening/reading basics	1	2025-10-28 07:01:37.127658+00

Bước 10. Tạo các course session:

```
WITH p AS (
  SELECT cp.id phase_id
  FROM course c
  JOIN course_phase cp ON cp.course_id = c.id AND cp.phase_number = 1
  WHERE c.code = 'CHI-HSK3-PREP-V1'
)
INSERT INTO course_session (phase_id, sequence_no, topic, student_task, skill_set,
created_at)
SELECT p.phase_id, v.sequence_no, v.topic, v.student_task, v.skill_set, NOW()
FROM p, (VALUES
  (1 , 'HSK3 - P1.S1: Greetings and self-introduction' , 'Workbook pp.1-3'
, ARRAY['listening','speaking']::skill_enum[]),
  (2 , 'HSK3 - P1.S2: Numbers and telling time'        , 'Workbook pp.4-6'
, ARRAY['listening','speaking']::skill_enum[]),
  (3 , 'HSK3 - P1.S3: Daily routines'                  , 'Dictation + drills'
, ARRAY['reading','writing']::skill_enum[]),
  (4 , 'HSK3 - P1.S4: Family and relationships'        , 'Sentence building'
, ARRAY['speaking','writing']::skill_enum[]),
  (5 , 'HSK3 - P1.S5: Food and ordering'               , 'Role-play + menu
reading'    , ARRAY['listening','speaking']::skill_enum[]),
  (6 , 'HSK3 - P1.S6: Shopping and prices'             , 'Price comparison task'
, ARRAY['reading','speaking']::skill_enum[]),
  (7 , 'HSK3 - P1.S7: Transportation and directions'   , 'Map task'
, ARRAY['listening','speaking']::skill_enum[]),
  (8 , 'HSK3 - P1.S8: Weather and seasons'             , 'Short paragraph writing'
, ARRAY['reading','writing']::skill_enum[]),
  (9 , 'HSK3 - P1.S9: Health and at the clinic'        , 'Dialogue practice'
, ARRAY['listening','speaking']::skill_enum[]),
  (10, 'HSK3 - P1.S10: Hobbies and leisure'            , 'Interview a partner'
, ARRAY['speaking']::skill_enum[]),
  (11, 'HSK3 - P1.S11: School and work'                , 'Note-taking from audio'
, ARRAY['listening','writing']::skill_enum[]),
  (12, 'HSK3 - P1.S12: Review + Mini-mock A'           , 'Mini-mock A + feedback'
, ARRAY['general']::skill_enum[])
) v(sequence_no, topic, student_task, skill_set)
ON CONFLICT (phase_id, sequence_no) DO NOTHING
RETURNING id, phase_id, sequence_no, topic, student_task, skill_set, created_at;
```

Results Chart Export Source Primary Database Role postgres Run CTRL

id	phase_id	sequence_no	topic	student_task	skill_set	created_at
43	10	1	HSK3 - P1.S1: Greetings and self-introduction	Workbook pp.1-3	{listening,speaking}	2025-10-28 09:26:10.64626+00
44	10	2	HSK3 - P1.S2: Numbers and telling time	Workbook pp.4-6	{listening,speaking}	2025-10-28 09:26:10.64626+00
45	10	3	HSK3 - P1.S3: Daily routines	Dictation + drills	{reading,writing}	2025-10-28 09:26:10.64626+00
46	10	4	HSK3 - P1.S4: Family and relationships	Sentence building	{speaking,writing}	2025-10-28 09:26:10.64626+00
47	10	5	HSK3 - P1.S5: Food and ordering	Role-play + menu reading	{listening,speaking}	2025-10-28 09:26:10.64626+00
48	10	6	HSK3 - P1.S6: Shopping and prices	Price comparison task	{reading,speaking}	2025-10-28 09:26:10.64626+00
49	10	7	HSK3 - P1.S7: Transportation and directions	Map task	{listening,speaking}	2025-10-28 09:26:10.64626+00
50	10	8	HSK3 - P1.S8: Weather and seasons	Short paragraph writing	{reading,writing}	2025-10-28 09:26:10.64626+00
51	10	9	HSK3 - P1.S9: Health and at the clinic	Dialogue practice	{listening,speaking}	2025-10-28 09:26:10.64626+00
52	10	10	HSK3 - P1.S10: Hobbies and leisure	Interview a partner	{speaking}	2025-10-28 09:26:10.64626+00

Bước 11. Check đã tạo đủ số session theo từng phase chưa

```
WITH x AS (
  SELECT cp.id phase_id,
         cp.phase_number,
         cp.duration_weeks,
         c.session_per_week,
         cp.duration_weeks * c.session_per_week expected_sessions,
         count(cs.id) created_sessions
  FROM course c
  JOIN course_phase cp ON cp.course_id = c.id
  LEFT JOIN course_session cs ON cs.phase_id = cp.id
  WHERE c.code = 'CHI-HSK3-PREP-V1'
  GROUP BY cp.id, cp.phase_number, cp.duration_weeks, c.session_per_week
)
SELECT phase_id, phase_number, expected_sessions, created_sessions,
       (created_sessions = expected_sessions) is_complete
FROM x
ORDER BY phase_number;
```

Results Chart Export Source Primary Database Role postgres Run CTRL

phase_id	phase_number	expected_sess	created_sessi	is_complete
10	1	12	12	true
11	2	12	12	true
12	3	12	12	true

Bước 12. Mapping CLOs - Course Session

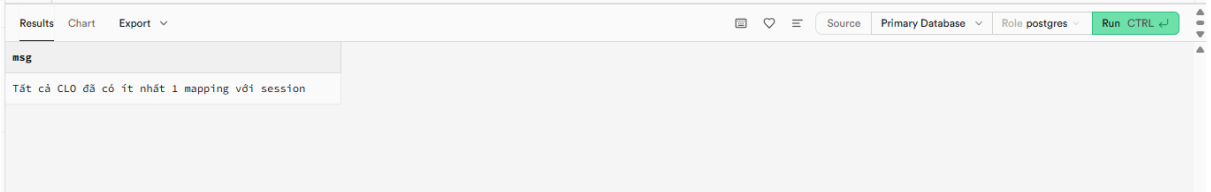
```
WITH pairs AS (
  SELECT * FROM (VALUES
    ('CLO1', 1, 1),
    ('CLO2', 1, 2),
    ('CLO4', 1, 3),
    ('CLO3', 1, 4),
    ('CLO1', 1, 5),
    ('CLO2', 1, 6),
    ('CLO4', 1, 7),
    ('CLO3', 1, 8),
```

```sql
    ('CLO1', 1, 9),
    ('CLO2', 1, 10),
    ('CLO4', 1, 11),
    ('CLO3', 1, 12),

    ('CLO1', 2, 1),
    ('CLO2', 2, 2),
    ('CLO4', 2, 3),
    ('CLO3', 2, 4),
    ('CLO1', 2, 5),
    ('CLO2', 2, 6),
    ('CLO4', 2, 7),
    ('CLO3', 2, 8),
    ('CLO1', 2, 9),
    ('CLO2', 2, 10),
    ('CLO4', 2, 11),
    ('CLO3', 2, 12),

    ('CLO1', 3, 1),
    ('CLO2', 3, 2),
    ('CLO4', 3, 3),
    ('CLO3', 3, 4),
    ('CLO1', 3, 5),
    ('CLO2', 3, 6),
    ('CLO4', 3, 7),
    ('CLO3', 3, 8),
    ('CLO1', 3, 9),
    ('CLO2', 3, 10),
    ('CLO4', 3, 11),
    ('CLO3', 3, 12)

  ) v(clo_code, phase_number, sequence_no)
),
ins AS (
  INSERT INTO course_session_clo_mapping (clo_id, course_session_id)
  SELECT clo.id, course_session.id
  FROM course
  JOIN course_phase ON course_phase.course_id = course.id
  JOIN course_session ON course_session.phase_id = course_phase.id
  JOIN clo ON clo.course_id = course.id
  JOIN pairs ON pairs.clo_code = clo.code
            AND pairs.phase_number = course_phase.phase_number
            AND pairs.sequence_no = course_session.sequence_no
  WHERE course.code = 'CHI-HSK3-PREP-V1'
```

```
  ON CONFLICT (clo_id, course_session_id) DO NOTHING
  RETURNING clo_id, course_session_id
)
SELECT
  clo.code clo_code,
  course.code course_code,
  course_phase.phase_number,
  course_session.sequence_no
FROM ins
JOIN clo ON clo.id = ins.clo_id
JOIN course_session ON course_session.id = ins.course_session_id
JOIN course_phase ON course_phase.id = course_session.phase_id
JOIN course ON course.id = course_phase.course_id
ORDER BY clo_code, course_phase.phase_number, course_session.sequence_no;
```

Bước 13. Check CLO mapping với các course session (còn CLO hay session chưa map không)

```
SELECT CASE
  WHEN EXISTS (
    SELECT 1
    FROM clo
    LEFT JOIN course_session_clo_mapping m ON m.clo_id = clo.id
    WHERE clo.course_id = (SELECT id FROM course WHERE code = 'CHI-HSK3-PREP-V1')
    GROUP BY clo.id
    HAVING COUNT(m.course_session_id) = 0
  )
  THEN 'Còn CLO chưa được mapping với session'
  ELSE 'Tất cả CLO đã có ít nhất 1 mapping với session'
END msg;
```



- Xem danh sách course session và CLO tương ứng

```
WITH base AS (
  SELECT
    c.id            AS course_id,
    c.code          AS course_code,
    cp.phase_number,
    cs.id           AS course_session_id,
    cs.sequence_no,
```

```
    cs.topic,
    cs.student_task,
    cs.skill_set
  FROM course c
  JOIN course_phase cp ON cp.course_id = c.id
  JOIN course_session cs ON cs.phase_id = cp.id
  WHERE c.code = 'CHI-HSK3-PREP-V1'
)
SELECT
  row_number() OVER (PARTITION BY base.course_id ORDER BY base.phase_number,
base.sequence_no) AS session_no,
  base.phase_number,
  base.sequence_no,
  base.topic,
  COALESCE(string_agg(DISTINCT clo.code, ', ' ORDER BY clo.code), '—')      AS
clo_list,         -- các CLO của buổi
  base.student_task,
  base.skill_set
FROM base
LEFT JOIN course_session_clo_mapping csm ON csm.course_session_id =
base.course_session_id
LEFT JOIN clo ON clo.id = csm.clo_id
GROUP BY
  base.course_id, base.phase_number, base.sequence_no,
  base.topic, base.student_task, base.skill_set
ORDER BY session_no;
```

Results Chart Export Source Primary Database Role postgres Run CTRL

session_no	phase_number	sequence_no	topic	clo_list	student_task	skill_set
1	1	1	HSK3 - P1.S1: Greetings and self-introduction	CLO1	Workbook pp.1-3	{listening,speaking}
2	1	2	HSK3 - P1.S2: Numbers and telling time	CLO2	Workbook pp.4-6	{listening,speaking}
3	1	3	HSK3 - P1.S3: Daily routines	CLO4	Dictation + drills	{reading,writing}
4	1	4	HSK3 - P1.S4: Family and relationships	CLO3	Sentence building	{speaking,writing}
5	1	5	HSK3 - P1.S5: Food and ordering	CLO1	Role-play + menu reading	{listening,speaking}
6	1	6	HSK3 - P1.S6: Shopping and prices	CLO2	Price comparison task	{reading,speaking}
7	1	7	HSK3 - P1.S7: Transportation and directions	CLO4	Map task	{listening,speaking}
8	1	8	HSK3 - P1.S8: Weather and seasons	CLO3	Short paragraph writing	{reading,writing}

Bước 14. Tạo materials gắn với course

```
WITH c AS (
  SELECT id course_id
  FROM course
  WHERE code = 'CHI-HSK3-PREP-V1'
)
INSERT INTO course_material
(course_id, phase_id, course_session_id, title, url, uploaded_by, created_at)
```

```sql
SELECT c.course_id, NULL, NULL,
       'General English A1 Course Syllabus',
       'https://drive.google.com/file/course1-syllabus',
       6, NOW()
FROM c
ON CONFLICT DO NOTHING
RETURNING id, course_id, phase_id, course_session_id, title, url, uploaded_by,
          (SELECT full_name FROM user_account u WHERE u.id =
course_material.uploaded_by) uploaded_by_name,
          created_at;
```

title	url	uploaded_by	uploaded_by_name	created_at
General English A1 Course Syllabus	https://drive.google.com/file/course1-syllabus	6	Nguyen Thi English Leader	2025-10-28 11:59:49.60147+00

- Xem danh sách material của khóa học

```sql
SELECT
  cm.id,
  c.code course_code,
  cm.title,
  cm.url,
  ua.full_name uploaded_by_name,
  cm.created_at,
  cm.updated_at
FROM course_material cm
JOIN course c ON c.id = cm.course_id
LEFT JOIN user_account ua ON ua.id = cm.uploaded_by
WHERE c.code = 'CHI-HSK3-PREP-V1'
ORDER BY cm.created_at;
```

Results Chart Export Source Primary Database Role postgres Run CTRL

id	course_code	title	url	uploaded_by_name	created_at
19	CHI-HSK3-PREP-V1	HSK Course Syllabus	https://drive.google.com/file/course1-syllabus	Nguyen Thi English Leader	2025-10-28 12:28:13.
20	CHI-HSK3-PREP-V1	HSK Course Listening Skill Syllabus	https://drive.google.com/file/course1-syllabus-listening	Nguyen Thi English Leader	2025-10-28 12:28:50.
21	CHI-HSK3-PREP-V1	HSK Course Listening Skill Slide	https://drive.google.com/file/course1-syllabus-slide	Nguyen Thi English Leader	2025-10-28 12:29:50.

Bước 15. Subject Leader define khung đánh giá khóa học

```sql
INSERT INTO course_assessment
  (course_id, name, kind, max_score, description, skills, created_at)
SELECT
  c.id,
  v.name,
  v.kind::assessment_kind_enum,
  v.max_score,
  v.description,
  v.skills,
  NOW()
FROM course c,
( VALUES
  ('Listening Quiz 1',  'quiz',    20.00, 'Quiz kĩ năng nghe theo chủ đề tuần',
ARRAY['listening']::skill_enum[]),
```

```
  ('Reading Quiz 1',     'quiz',     25.00, 'Quiz kĩ năng đọc đoạn ngắn + từ vựng',
ARRAY['reading']::skill_enum[]),
  ('Speaking Test',      'oral',     30.00, 'Kiểm tra nói: self-intro, role-play
ngắn',     ARRAY['speaking']::skill_enum[]),
  ('Midterm Exam',       'midterm', 50.00, 'Giữa kỳ bao quát Phase 1–2',
ARRAY['general','reading','writing']::skill_enum[]),
  ('Final Exam HSK3',    'final',  100.00, 'Cuối kỳ 4 kĩ năng theo format HSK3',
ARRAY['general','reading','writing','listening','speaking']::skill_enum[]),
  ('Class Participation','other',  20.00, 'Điểm chuyên cần/nhiệm vụ hằng tuần',
ARRAY['general']::skill_enum[])
) AS v(name, kind, max_score, description, skills)
WHERE c.code = 'CHI-HSK3-PREP-V1'
  AND NOT EXISTS (
    SELECT 1
    FROM course_assessment ca
    WHERE ca.course_id = c.id
      AND LOWER(ca.name) = LOWER(v.name)
  )
RETURNING id, course_id, name, kind, max_score, description, skills, created_at;
```

Results Chart Export Source Primary Database Role postgres Run select

id	course_id	name	kind	max_score	description	skills
19	13	Listening Quiz 1	quiz	20.00	Quiz kĩ năng nghe theo chủ đề tuần	{listening}
20	13	Final Exam HSK3	final	100.00	Cuối kỳ 4 kĩ năng theo format HSK3	{general,reading,writing,listening,speaking}
21	13	Speaking Test	oral	30.00	Kiểm tra nói: self-intro, role-play ngắn	{speaking}
22	13	Reading Quiz 1	quiz	25.00	Quiz kĩ năng đọc đoạn ngắn + từ vựng	{reading}
23	13	Midterm Exam	midterm	50.00	Giữa kỳ bao quát Phase 1–2	{general,reading,writing}
24	13	Class Participation	other	20.00	Điểm chuyên cần/nhiệm vụ hằng tuần	{general}

2. Map Assessment ↔ CLO

```
INSERT INTO assessment_clo_mapping (
  course_id, assessment_template_id, clo_id, contribution_pct, created_by, created_at
) VALUES (
  :course_id, :assessment_template_id, :clo_id, :contribution_pct, :sl_user_id, NOW()
);
```

- Subject Leader submit Course

```
UPDATE course
SET status       = 'submitted',
    submitted_at = NOW(),
    updated_by   = :sl_user_id,
    updated_at   = NOW()
WHERE id = :course_id
  AND status = 'draft';
```

- Manager approve/reject

1. Approve

```
UPDATE course
SET status              = 'approved',
    approved_by_manager = :manager_user_id,
    approved_at         = NOW(),
    updated_by          = :manager_user_id,
    updated_at          = NOW()
WHERE id = :course_id
  AND status = 'submitted';
```

2. Manager reject (course sẽ quay lại status “draft”)

```
UPDATE course
SET status           = 'draft',
    rejection_reason = :reason,
    updated_by       = :manager_user_id,
    updated_at       = NOW()
WHERE id = :course_id
  AND status = 'submitted';
```